Ejercicio 2

regula la velocidad de crec.

$$x_{n+1} = x_n + a\, x_n \underbrace{(x_n - b)}_{\text{efecto Allee}} \underbrace{(1 - x_n)}_{\text{Como en modelo logístico}}$$

$0 < b < 1$
$0 < a$

$\Rightarrow$ es un modelo logístico con efecto Allee.

1. Puntos de Equilibrio + Estabilidad

$$x^* = \underbrace{x^* + a x^* (x^* - b)(1 - x^*)}_{f(x^*)} \Leftrightarrow a x^* (x^* - b)(1 - x^*) = 0 \Leftrightarrow \left\{ \begin{array}{l} x^* = 0 \\ x^* = b \\ x^* = 1 \end{array} \right.$$

Puntos fijos

$$f'(x) = 1 + a\left[(x-b)(1-x) + x(1-x) - x(x-b) \right]$$

$\boxed{x^* = 0}$

$$f'(0) = 1 + a[-b] = 1 - ab$$

0 estable si $|f'(0)| = |1 - ba| < 1 \Rightarrow \begin{array}{l} 1 - ba < 1 \Leftrightarrow ba > 0 \\ 1 - ba > -1 \Leftrightarrow ba < 2 \end{array} \Big\}\ ba \in]0, 2[$

0 inestable si $|\ \ | > 1 \Rightarrow ba < 0$ ó $ba > 2$

¿ $ab = 0$ ó $ab = 2$?

No es posible $a > 0$, $b > 0$

$(a = 2/b) \Rightarrow f'(0) = 1 - ab = 1 - 2 = -1$

$$f(x) = x + a x (x - b)(1 - x)$$

$$f'(x) = 1 + a\left[(x-b)(1-x) + x(1-x) - x(x-b)\right] = 1 + a\left(-3x^2 + 2(1+b)x - b\right)$$

$$f''(x) = a\left(-6x + 2(1+b)\right)$$

$$f'''(x) = -6a$$

$$f''(0) = a\left(2(1+b)\right) = \frac{2}{b}\left(2(1+b)\right) = \frac{4b+4}{b}$$

$$f'''(0) = -6a = \frac{-12}{b}$$

$$2 f'''(0) + 3\left(f''(0)\right)^2 = \frac{-24}{b} + 3\left(\frac{(4b+4)^2}{b^2}\right) = \frac{-24}{b} + 3\,\frac{16b^2 + 32b + 16}{b^2} = \frac{48b^2 + 72b + 16}{b^2} > 0$$

$\Downarrow$ A. Estable localmente

Entonces, $x^* = 0$ es A.E. loc $\Leftrightarrow ba \in]0, 2]$

$\boxed{x^* = b}$ $f'(b) = 1 + a[\,b(\underset{>0}{1 - b})\,] > 1 \Rightarrow$ Inestable siempre.

$\boxed{x^* = 1}$ $f'(1) = 1 + a[-1(1 - b)]$

Si $|f'(1)| < 1 \Rightarrow x = 1$ estable ¿

siempre

$\Leftrightarrow a(1 - b) > 0\ ]$

$$\left.\begin{array}{l} 1 - a(1-b) < 1 \iff a(1-b) > 0 \\ 1 - a(1-b) > -1 \iff \underline{a(1-b) < 2} \end{array}\right\}$$

- $¿\, a(1-b) = 0\, ? \equiv a = 0$ ó $b = 1$ $\boxed{\text{No}}$
- $¿\, a(1-b) = 2\, ? \Rightarrow a = \frac{2}{1-b}$

$$f''(x) = a(-6x + 2(1+b))$$
$$f'''(x) = -6a$$

$$f'(1) = 1 - 2 = -1$$

$$f''(1) = \frac{2}{1-b}(-6 + 2(1+b)) = \frac{-12 + 4 + 4b}{1-b} = \frac{4b - 8}{1-b}$$

$$f'''(1) = -6 \cdot \frac{2}{1-b} = \frac{-12}{1-b}$$

$$2f'''(1) + 3(f''(1))^2 = \frac{-24}{1-b} + 3 \cdot \frac{16b^2 - 64b + 64}{1-b} = \frac{48b^2 - 192 + 168}{\underset{>0}{1-b}} > 0 \iff$$

$$\iff 24b^2 - 96b + 84 > 0 \iff 12b^2 - 48 + 42 > 0 \iff 6b^2 - 24b + 21 > 0$$

$$\Updownarrow$$

Puntos de corte:

$$b = \frac{24 \pm \sqrt{24^2 - 4 \cdot 6 \cdot 21}}{2 \cdot 6} = \frac{24 \pm 6\sqrt{2}}{12} < \begin{array}{l} \frac{24+6\sqrt{2}}{12} \simeq 2,7071 \\ \frac{24-6\sqrt{2}}{12} \simeq 1,2929 \end{array}$$

$$\underline{b \in \left]\frac{24-6\sqrt{2}}{12}, \frac{24+6\sqrt{2}}{12}\right[}$$

$$\Downarrow$$

$\underline{x = 1 \text{ estable}}$

Resumen:

$$x=1 \text{ A.E. loc} \iff \begin{cases} a(1-b) < 2 \\ \text{ó} \\ a(1-b) = 2 \text{ con } b \in \left]\frac{24-6\sqrt{2}}{12}, \frac{24+6\sqrt{2}}{12}\right[\end{cases}$$

2. **2-ciclos** → Demasiado jodido numéricamente en general, en todo caso en la simulación dando valores se puede ver gráficamente.